

Requirements Table

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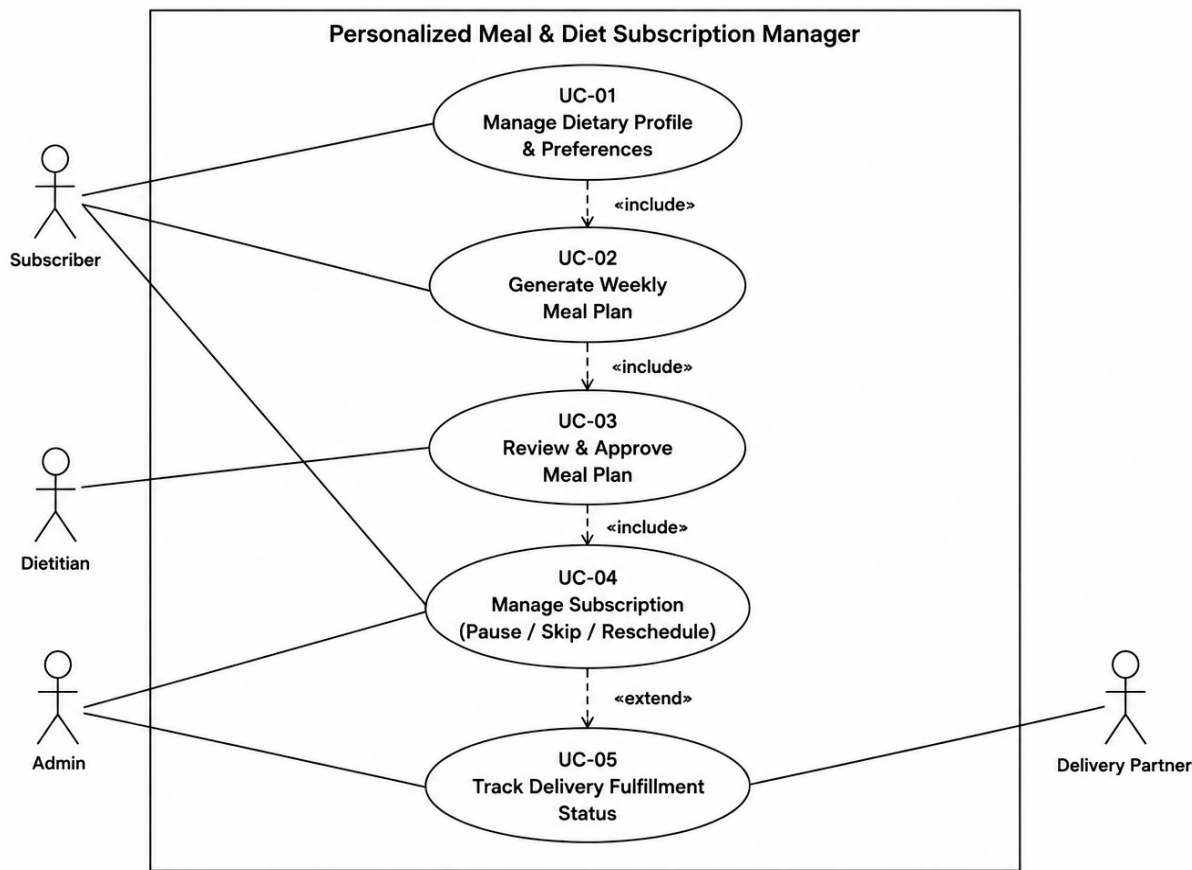
SRN : PES1UG24AM394

Personalized Meal & Diet Subscription Manager

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate weekly meal plans strictly adhering to user macro-nutrient constraints and allergen exclusion lists.	High	Pass: Generated menu contains zero excluded allergen ingredients. Fail: Any excluded allergen ingredient is present.	Ensures that generated meals are safe and personalized to the subscriber.
FR-002	Functional	The system shall allow the Subscriber to enter and update dietary constraints, allergies, diabetic limits and calorie goals.	High	Pass: Updated dietary information is saved and displayed in the subscriber profile. Fail: Updated information is not saved.	Provides the information required for personalized meal planning.
FR-003	Functional	The system shall allow the Dietitian to review and approve the generated weekly meal plan before dispatch.	High	Pass: Dietitian approves the plan and its status changes to Approved. Fail: The plan cannot be approved.	Allows professional dietary review before meal dispatch.
FR-004	Functional	The system shall allow the Subscriber to pause, skip or reschedule a meal subscription before dispatch.	Medium	Pass: A valid pause, skip or reschedule request updates the subscription schedule. Fail: The schedule remains unchanged.	Provides flexibility for subscription schedule management.
FR-005	Functional	The system shall record and display the daily delivery fulfillment status for each scheduled meal.	High	Pass: A completed delivery is recorded with Delivered status. Fail: The delivery status is not updated.	Allows the system to track daily meal delivery fulfillment.
NFR-001	Nonfunctional — Performance & Security	The subscription engine shall support schedule modifications (pause, skip, reschedule) up to 12 hours before dispatch.	High	Pass: Valid modifications made at least 12 hours before dispatch are accepted. Fail: A valid modification before the cutoff is rejected.	Supports the 12-hour schedule modification constraint specified in the problem statement.
NFR-002	Nonfunctional — Security	The system shall protect subscriber dietary and allergy information from unauthorized access.	High	Pass: Unauthorized users cannot view or modify restricted dietary information. Fail: An unauthorized user can access or modify the information.	Protects sensitive subscriber dietary and allergy information.

UML Use-Case Diagram: Personalized Meal & Diet Subscription Manager

UML Use-Case Diagram - Personalized Meal & Diet Subscription Manager



Use-Case Flow

Use Case: Generate Weekly Meal Plan

Primary Actor: Subscriber

Supporting Actor: Dietitian

Preconditions

- The Subscriber has an active meal subscription.
- The Subscriber's dietary profile is available.
- Allergies and dietary constraints are recorded.
- Calorie goals and macro-nutrient constraints are available.

Postconditions

- A validated weekly meal plan is generated.
- The approved meal plan is stored for the Subscriber.
- The meal plan is ready for scheduled fulfillment.

Main Success Scenario

1. Subscriber accesses the meal subscription system.
2. System retrieves the Subscriber's dietary profile.
3. System retrieves allergies, diabetic limits, calorie goals and macro-nutrient constraints.
4. System validates the dietary constraints.
5. System generates a weekly meal plan.
6. System checks the generated meals against the allergen exclusion list.
7. System verifies that the meal plan satisfies the specified nutritional constraints.
8. System displays the generated weekly meal plan.
9. Dietitian reviews the generated meal plan.
10. Dietitian approves the meal plan.
11. System stores the approved weekly meal plan.
12. Use case ends successfully.

Alternate Flow — Generated Meal Violates a Constraint

- 5a1. System detects that a generated meal contains an excluded allergen or violates a dietary constraint.
- 5a2. System rejects the meal.
- 5a3. System generates a replacement meal.
- 5a4. System validates the replacement meal against the dietary constraints.
- 5a5. If the replacement satisfies all constraints, the system adds it to the weekly meal plan and continues from Step 8 of the Main Success Scenario.